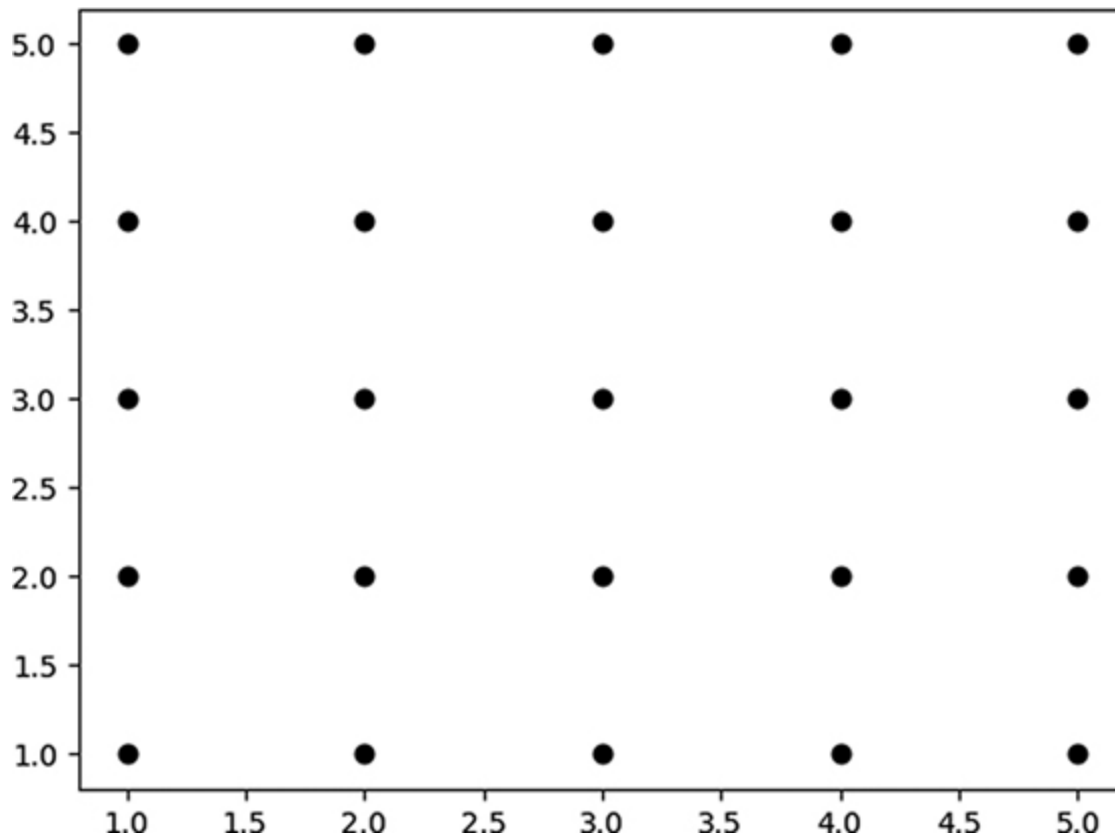




*[Figure 10-1](#): A lattice, which we can use as a game board for dots and boxes*

The lattice is usually a rectangle but can be any shape. Two players play against each other, taking turns. On each turn, a player is allowed to draw a line segment that connects two adjacent points in the lattice. If they use different colors to draw their line segments, we can see who has drawn what, though that's not required. As they proceed through the game, line segments fill the lattice until every possible segment connecting adjacent points is drawn. You can see an example game in progress in [Figure 10-2](#).

A player's goal in dots and boxes is to draw line segments that complete squares. In [Figure 10-2](#), you can see that in the bottom left of the game board, one square has been completed. Whichever player drew the line segment that completed that square will have earned one point from doing so. In the top-right section, you can see that three sides of another square have been drawn. It's player one's turn, and if they use their turn to draw a line segment between (4,4) and (4,3), they'll earn one point for that. If instead they draw another line segment, like a line segment from (4,1) to (5,1), then they'll give player two a chance to finish the square